

NeuroTechX

NeuroTechX: Building a Community

Our work, our contribution, and the next neurotech founders

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Japan BMI/BCI Ecosystem Kick-off
Coral Capital, Tokyo / Hybrid
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What is NeuroTechX?

A nonprofit community

Bringing together people interested in neuroscience and technology.

Local beginnings, international connections

Founded in Montréal in 2015; connected through city chapters, student groups, and open projects.

Three pillars

Community, education, and professional development.

[NeuroTechX: Our Story](#)

What do we do?

Bring people together

Chapter meetings, hacknights, student communities, and hackathons.

Make learning accessible

The NeuroTech Primer, workshops, webinars, and practical experiments.

Build and share tools

EEG-ExPy, MOABB, and contributions across the open neurotechnology ecosystem.

NeuroTechX: education and developer resources; Morgan's original presentation

NeuroTechSF: the weekly hacknight

A recurring place to meet

Newcomers welcome alongside people already working on projects.

In-person and remote participation

Our chapter's format includes online participation, livestreaming, and recordings.

Something to work on together

Hardware for hacking, experiments, software, and help from peers.

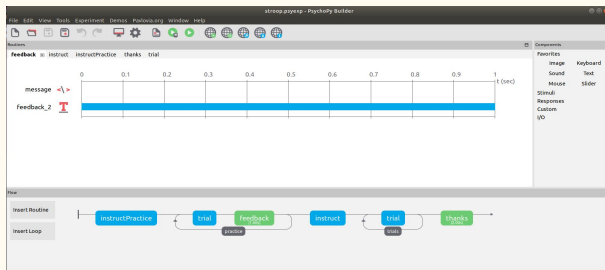
NeuroTechSF events; format described in [Morgan's original slides](#)

EEG-ExPy: a shared experiment

EEG experiments in Python

A project people can run, inspect, and help improve.

Experiment design connects with software and data collection.



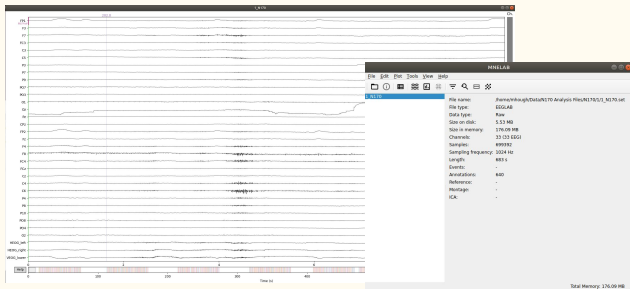
EEG-ExPy; experiment illustration from Morgan's original slide 10

Citizen science: from recording to analysis

Participation can include the experiment itself.

Our examples include N170 experiments and resting-state recordings.

Looking at the signal is part of learning how to collect it.



Morgan's original presentation: citizen science and EEG analysis

MOABB: making BCI comparisons reproducible

Mother of All BCI Benchmarks

An open framework for comparing algorithms on public EEG datasets.

A concrete research contribution

The 2024 benchmark study evaluated 30 machine-learning pipelines across 36 datasets, covering motor imagery, P300, and SSVEP.

Shared implementations and evaluation methods give other researchers a starting point they can inspect and reproduce.

[Chevallier et al., 2024, arXiv:2404.15319](#)

Education: a way into the field

The NeuroTech Primer

An introduction to neurotechnology, including recording and stimulation.

Workshops and webinars

Opportunities to learn methods and hear from people working in the field.

Learning through contribution

Developer meetings and practical projects alongside written resources.

NeuroTechX: The NeuroTech Primer, learning resources, and developer meetings

What these activities contribute

Access to people

Meetings and events create opportunities to ask questions and find collaborators.

Access to practice

Shared experiments and workshops give people something concrete to try.

Work others can reuse

Code, benchmarks, recordings, and learning materials remain available beyond a meeting.

Synthesis of the activities and resources shown in this talk

The global hackathon: 2022

October 29–30, 2022

Global online participation, with local chapter hosting.
Free and open to the public.

Five listed in-person locations

Paris, Barcelona, Belgrade, San Francisco, and Munich.

Three ways to participate

Hands-on technology; an EEG data challenge;
design, strategy, and other disciplines.

neurotechx.com/hackathon2022

The global hackathon: 2023

December 2–3, 2023

The second annual global event used a hybrid format.
Free and open to the public.

Eight listed in-person locations

Paris, Vienna, Zurich, Delhi, Kharagpur,
Brasília, Buenos Aires, and London.

Four tracks

Technology and hardware; data and algorithms;
design and strategy; a newly added ethics track.

neurotechx.com/hackathon2023

Global NeuroHack 2026

April 10–12, 2026

Frontier Tower, San Francisco

A recent event we hosted, with NeuroTechX helping organize and support the hackathon.

Three tracks:

- Blue-Sky Prototyping
- Non-Implantable Speech Decoding
- Maximizing Bit Rate

The program included workshops, mentor check-ins, and project presentations.

global-neurohack.github.io

What the hackathon history shows

Different ways to join

The 2022 and 2023 programs included technical work, data, and design.

Different places to participate

The earlier global programs combined online participation with local hosts.

Different questions to work on

The 2023 program added ethics; the 2026 tracks included speech decoding and BCI games.

2022 and 2023 event programs; Global NeuroHack 2026 program

Founders can build on work already shared

Learn the field

Use educational resources and community conversations to develop a working understanding.

Explore an idea

Use open experiments and tools to investigate a technical question.

Test the claim

Use reproducible baselines to ask what an approach actually adds.

Founder-facing implications of the resources shown; MOABB and EEG-ExPy

Where could the next founder contribute?

Tools for people doing neurotechnology

Recording workflows, experiments, analysis, and reproducibility.

Applications built around a specific need

The recent hackathon explored speech interfaces, BCI games, and open-ended prototypes.

The practical questions

Who needs this? What works today? What evidence is still missing?

Global NeuroHack 2026 tracks; application questions for discussion

A founder can also be a community contributor

Bring a question people can work on

A technical problem, an experiment, or a result to examine together.

Share something useful

Documentation, a reproducible example, a workshop, or time helping another builder.

Build relationships through the work

Meet people by learning, making, and contributing alongside them.

What can we build and learn together?

Join the community

neurotechx.org/community

Explore the work

github.com/NeuroTechX

Find a NeuroTechSF event

lu.ma/neurotechsf

Morgan Hough

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Sources

NeuroTechX and its activities

Organization history Education and developer resources

Projects

EEG-ExPy MOABB 2024 MOABB benchmark study

Hackathon history

2022 2023 Global NeuroHack 2026

Firsthand presentation material

Morgan's original NeuroTechX presentation